Year 2 Computing – Semester 3
COMP H2031: OOAD
Object Oriented Analysis & Design

Academic term	**2024/25**
Year of study	**Year 2**
Semester	**SEMESTER ONE**
Date of examination	
Time of examination	**95minutes + 5 minutes to upload = total 100 minutes**

COMP H2031: Object Oriented Analysis & Design

Java Coding Lab Test 1

SAMPLE PAPER – GIVEN TO STUDENS Oct 2024

Internal Examiner(s)	**Orla McMahon, Krzysztof Szczurowski**

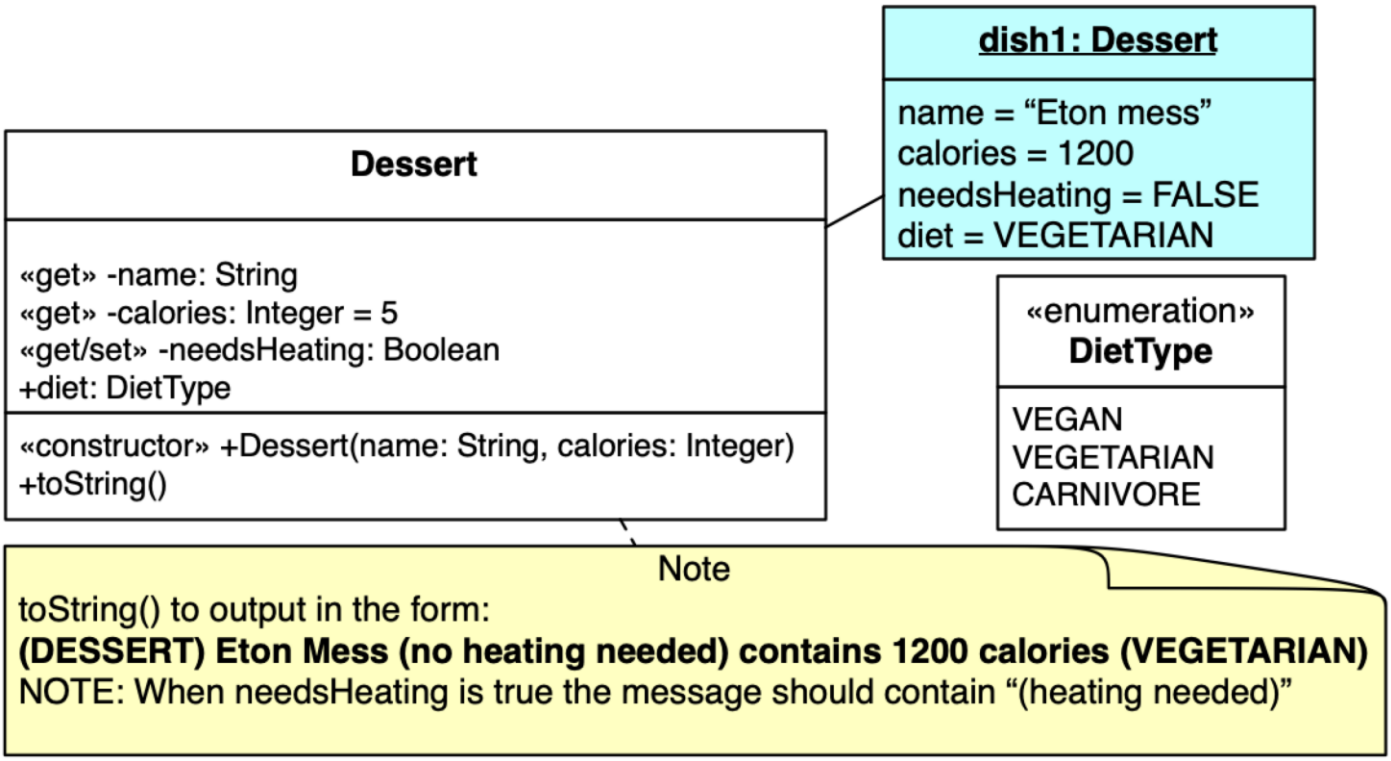
Instructions to candidates:

1.	**This is an individual test – all work should be your own. During the test you should NOT communicate with anyone apart from the lecturer (excepting medical / emergency circumstances).**
2.	**You must attend the test for your timetabled lab group. If you are not on the list for a group you will not be permitted to take the test.**
3.	**Bring physical photo ID: e.g. student card / driving licence / passport Place this beside you during the test.**
4.	**Use of headphones, phones, tablets, email etc. is NOT PERMITTED.**
6.	**This is an open-book test for paper-based materials. Use of GEN AI is not permitted**
7.	**The test is marked out of 100. Answer ALL questions.**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO

Question 1: Objects, enumeration & objects *Total: 40 marks*

Submit your answers to this question in folder: **question1/**



1a. Write Java files to implement the classes in the diagram above.

1b. Write a **main(…)** method in class **Main** to do the following:

- create and print out the details of the object(s) in the diagram.

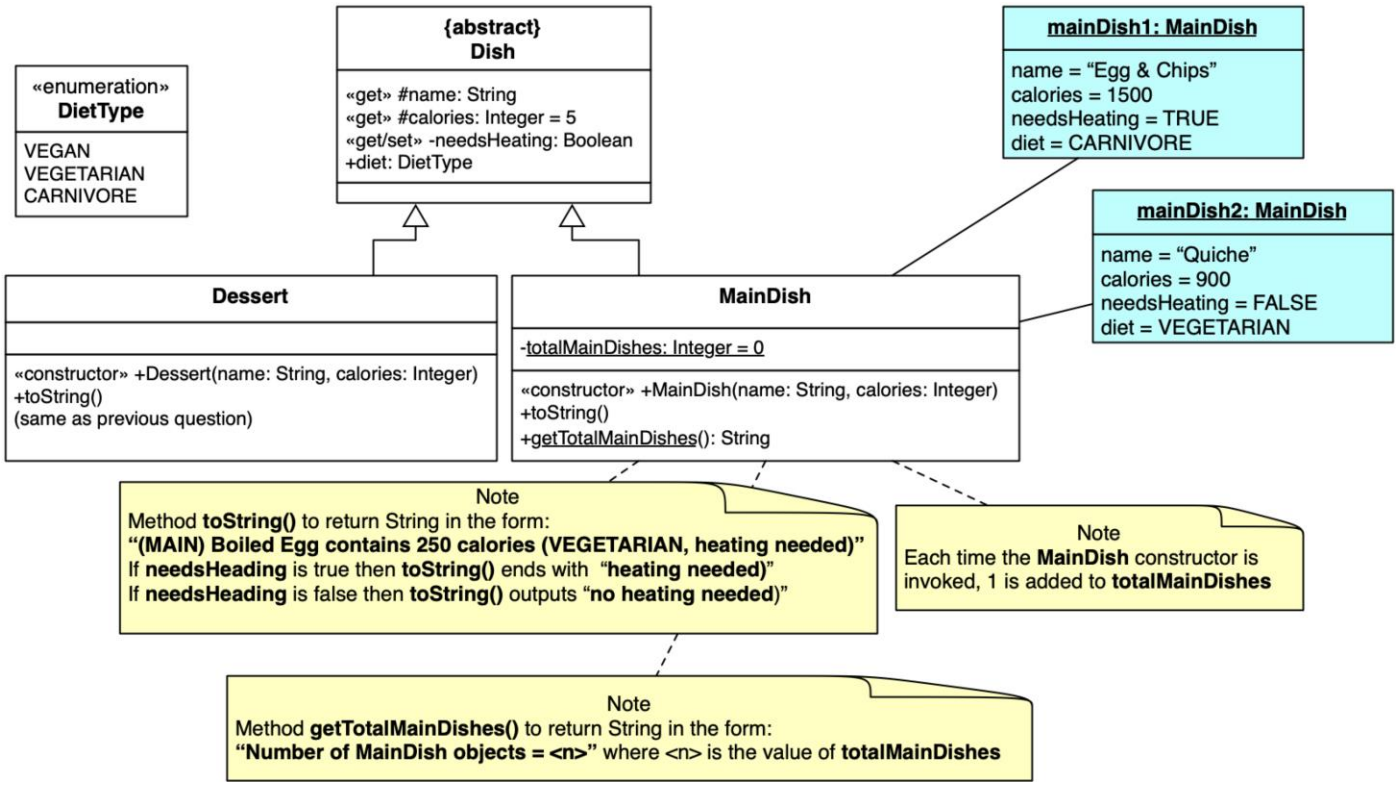
Your program output should look as follows:

```
$ java Main

(DESSERT) Eton Mess (no heating needed) contains 1200 calories (VEGETARIAN)
```

Question 2: Inheritance, protected & static *Total: 60 marks*

Copy your question 1 folder, naming the copy: **question2/**. Adapt your code to answer this question.



2a. Write Java files to implement the classes in the diagram above.

2b. Write a **main(…)** method in class **Main** to do the following:

- Print out the String returned by **MainDish** method **getTotalMainDishes()**
- Create the first **MainDish** object from the diagram, and print out its details
- Print out the String returned by **MainDish** method **getTotalMainDishes()**
- Create the second **MainDish** object from the diagram, and print out its details
- Print out the String returned by **MainDish** method **getTotalMainDishes()**

Your program output should look as follows:

```
$ java Main

Number of MainDish objects = 0

(MAIN) Egg & Chips contains 1500 calories (CARNIVORE, heating needed)

Number of MainDish objects = 1

(MAIN) Quiche contains 900 calories (VEGETARIAN, no heating needed)

Number of MainDish objects = 2
```